



The Electrochemical Behavior of Alkali and Alkaline Earth Metals in Nonaqueous Battery Systems—The Solid Electrolyte Interphase Model

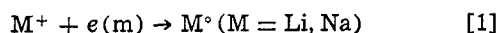
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ABSTRACT

It is suggested that in practical nonaqueous battery systems the alkali and alkaline earth metals are always covered by a surface layer which is instantly formed by the reaction of the metal with the electrolyte. This layer, which acts as an interphase between the metal and the solution, has the properties of a solid electrolyte. The corrosion rate of the metal, the mechanism of the deposition-dissolution process, the kinetic parameters, the quality of the metal deposit, and the half-cell potential depend on the character of the solid electrolyte interphase (SEI).

The electrochemical behavior of the alkali metals in nonaqueous systems was extensively studied in the last 15 years, mainly in connection with high energy battery systems. This work is summarized in Ref. (1-4). The electrode kinetics of the alkali metals in these systems were treated and analyzed following the Butler-Volmer equation. It was generally assumed (1-4) that the rate-determining step (rds) is the transfer of electrons from the metal to the ions in the solution. Some experimental results indicate that this assumption is oversimplified and the real situation is much more complex. For example, the anodic transfer coefficient (α_a) measured for the reaction



was in many cases 0.45-0.13 (1-5). In one case, α_a , (for reaction [1]) was found to be insensitive to the concentration of Li^+ ions in the solution (7) while in another case (6) it was extremely sensitive to the concentration of Na^+ in the solution. The most complex behavior of these systems is demonstrated in Table I. This table contains six similar nonaqueous systems. However, when these systems are electrolyzed, the first three yield aluminum as expected from thermodynamic considerations, while the other three yield lithium or magnesium in apparent contradiction to thermodynamics. Each of the first three systems is used as a bath for the electroplating of aluminum. The surfaces of the electrodes immersed in these systems are in direct contact with the solution either during electrolysis or under open-circuit conditions. The last three systems are examples of nonaqueous battery (NAB) systems. In these systems the electrodeposited metal is the more reactive one.

Recently, it was concluded by Dey (13), and Rauh and Brummer (14) that in some NAB systems the lithium metal is covered by a Li^+ conducting film. A more detailed model for such a film has been described for

the magnesium electrode in thionyl chloride solutions (12).

In this paper it will be emphasized that the electrode kinetics and the deposition-dissolution mechanism of the alkali and alkaline earth metals in NAB systems are entirely different from those characteristic of other nonaqueous or aqueous electrochemical systems. A model will be proposed aimed at explaining the complex and unique behavior of NAB systems.

The SEI model.—It is suggested that in practical NAB systems the alkali and alkaline earth metals are always covered by a surface layer at least 15-25 Å thick (12). This layer is formed instantly by the contact of the metal with the solution. The thickness of this freshly formed layer is determined by the electron tunneling range. This layer consists of some insoluble products of the reaction of the metal with the solution. It acts as an interphase between the metal and the solution and has the properties of solid electrolyte, through which electrons are not allowed to pass. Therefore, it is called "Solid Electrolyte Interphase (SEI)."

The existence of this electronic insulating layer on the anodes in NAB systems is the principle difference between NAB and aqueous-like systems such as, Examples 1-3 in Table I. It should be emphasized that

Table I. Electrolysis of nonaqueous systems

No.	Electrolyte	Solvent	Electrodeposited metal	Ref.
1	$\text{AlBr}_3 + \text{LiBr}$	Toluene	Al	(8)
2	$\text{AlCl}_3 + \text{LiCl} + \text{LiAlH}_4$	Diethyl ether	Al	(9)
3	$\text{AlCl}_3 + \text{LiCl} + \text{LiAlH}_4$	THF	Al	(10)
4	$\text{AlCl}_3 + \text{LiCl}$	Thionyl chloride	Li	(11)
5	$\text{AlCl}_3 + \text{LiCl}$	Propylene carbonate	Li	(1)
6	$\text{FeCl}_3 + \text{MgCl}_2$	Thionyl chloride	Mg	(12)

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in NAB systems (according to this suggestion) there is never a direct and free contact between the metal and the solution. Thus, these batteries will be called SEI batteries. The electrochemical behavior of SEI electrodes will be governed by the properties of the SEI. The following properties of the SEI should be considered: its morphology (compact or porous, size of the crystals); its thickness (L); the type and the concentration of the lattice defects; the transference numbers of electrons (t_e), cationic defect (t_+), anionic defect (t_-); the mobility of these defects. The potential difference $\Delta\phi_{M/sol}$ contains three components (12)

$$\Delta\phi_{M/sol} = \Delta\phi_{M/SE} + \Delta\phi_{SE} + \Delta\phi_{SE/sol} \quad [2]$$

the total overpotential contains similar terms.

The simplest and most important case for NAB systems is a SEI electrode having $t_+ = 1$ and $t_e = 0$, i.e., the SEI is a pure cationic conductor. In this case, the electrode will be reversible for its own cations and thus will obey the Nernst law. The dissolution-deposition process for SEI electrode having Schottky lattice defects is schematically described in Fig. 1. It has three steps: (i) on cathodic polarization a metallic cation sheds its solvent molecules, crosses the solution/SE interface and enters inside a vacancy in the SEI; (ii) it migrates through the bulk of the SEI by Schottky vacancies mechanism; (iii) finally, it reaches the metallic surface and accepts an electron from the metal and becomes a member of the metallic lattice. On anodic polarization the metallic cation moves in

the opposite direction. The dissolution-deposition process in a real SEI electrode is more complicated as some of the following phenomena can be involved: crystallization; grain boundary effect; shorts through the SEI during prolonged deposition; breakdown and repair of the SEI during prolonged dissolution; adatoms formation in the M/SE interface; and adsorbed ions in the SE/solution interface. Some of these phenomena will be discussed later.

It is possible to electrodeposit alkali or alkaline earth metals (12) from NAB systems on an inert cathode. However, in these cases the metal should start to deposit only after the electrode surface is completely covered by a passivating layer which acts as a SEI through which the electrons cannot pass (12).

Kinetics of the SEI electrode.—As it was mentioned the most important and simple case is $t_+ = 1$ ($t_-, t_e = 0$). In this case, the rds may be any one of the three steps described above. For highly conductive SEI at small thickness the charge transfer reaction at one of the SEI interfaces may be the rds. However, when the SEI is thick enough the migration of ions through it may be the rds. Moreover, it is known that the ionic conductivity of pure solid crystals is very low at room temperature (15), a fact which strengthens the last assumption. In this case it is possible to use Eq. [3] describing the migration of ions in a solid crystal under external field (16). It is assumed that L is larger than the space charge lengths in the SEI, and there is no change in the concentration or mobility of the mobile lattice defects through the SEI

$$i = 4ZFan_+ \nu \exp(-W/RT) \sinh(aZFE/RT) \quad [3]$$

a = half-jump distance, ν = vibration frequency of the ion in the crystal, n_+ = concentration of the lattice cationic defect, Z = the valance of the mobile ion, W = a barrier energy for jumping, E = electric field. The assumption that the migration of the cation in the SEI is the rds means that the largest fraction of the electrode overpotential (η) will develop on the SEI, i.e.

$$\eta = EL \quad [4]$$

On the substitution of Eq. [4] in Eq. [3], one gets

$$i = 4ZFan_+ \nu \exp(-W/RT) \sinh(aZF\eta/RTL) \quad [5]$$

At $\eta = 0$, the forward and backward ionic current through the SEI are equal and the ionic exchange current density (i_0') can be defined

$$i_0' = \vec{i} = \overleftarrow{i} \quad [6]$$

where i_0' is given by (7)

$$i_0' = 2ZFavn_+ \exp(-W/RT) \quad [7]$$

and Eq. [5] becomes

$$i = 2i_0' \sinh(aZF\eta/RTL) \quad [8]$$

For high field conditions ($aZF\eta > RTL$) a Tafel-like equation is obtained

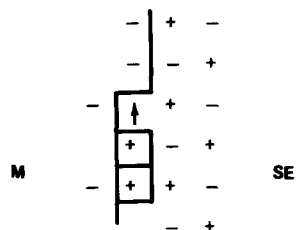
$$i = i_0' \exp(aZF\eta/RTL) \quad [9]$$

and the Tafel slope is

$$b = \frac{2.3 RTL}{aZF} \quad [10]$$

That means that one cannot expect to get a sole Tafel slope when measuring the kinetics of the alkali and the alkaline earth metals in NAB systems. The minimum value of b can roughly be estimated using Eq. [11]. For $z = 1$, $a = 3\text{\AA}$, $L = 20\text{\AA}$ one gets b 400 mV

$$b(\text{mV}) \approx \frac{60L}{aZ} \quad [11]$$



MIGRATION IN SE ($\Delta\phi_{SE}$)

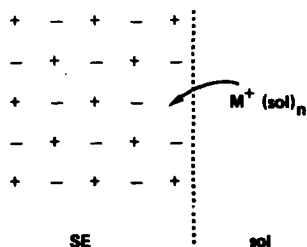
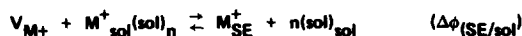
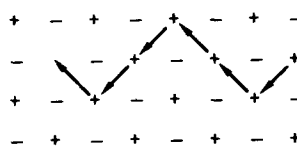


Fig. 1. A schematic description of the deposition-dissolution process (see text).

However, Tafel slopes of several volts can be measured for a thicker SEI electrode.

For low field conditions, Eq. [5] reduces to Ohm's law

$$i = \frac{4a^2Z^2F^2n_+ \exp(-W/RT)}{RT} \cdot \frac{\eta}{L} \quad [12]$$

or

$$i = \frac{1}{\rho} \cdot \frac{\eta}{L} \quad [13]$$

where ρ is the resistivity of the SEI. The reaction resistance of an electrode is defined as

$$R_r \equiv \left(\frac{\eta}{i} \right)_{\eta \rightarrow 0} \quad [14]$$

thus

$$R_r = \rho L \quad [15]$$

The relation between i_0' and ρ is given by

$$i_0' = \frac{RT}{2\rho aZF} \quad [16]$$

where

$$\rho = \frac{RT}{4a^2Z^2F^2n_+ \exp(-W/RT)} \quad [17]$$

The reaction resistance of SEI electrode increases linearly with the thickness of the SEI (Eq. [15]). Thus no sole value of R_r can be measured in these NAB systems.

Another interesting aspect is the way the concentration of the cation in the solution affects the reaction resistance. In the case of regular electrode kinetics regarding an electron transfer reaction the concentration effect is given by (18)

$$i_0(CT) = \text{const} \cdot C_M^{\alpha_a} \quad [18a]$$

or

$$R_r = \text{const} \cdot C_M^{-\alpha_a} \quad [18b]$$

where α_a is the anodic transfer coefficient and i_0 is the exchange current density for the electron charge transfer (CT) reaction.

However, in the case of SEI electrodes two extreme cases can be identified: (i) $L \gg$ space charge length and the rds is migration through the SEI. In this case the concentration of the cation in the solution is not expected to affect R_r and the calculated apparent α_a should be close to zero; (ii) $L \approx$ space charge length. In this case the concentration of the ions in the solution may affect the concentration of the defects in the SEI and thus may affect R_r , i_0' , and ρ . It can be expected that the apparent transfer coefficient as defined in Eq. [18b] will be decreased to zero as L increases. The direction of this effect depends on the type of the lattice defects and on the pzc of the SEI.

Literature examples.—The double-layer capacitance in nonaqueous solutions has values of 10–30 $\mu\text{F cm}^{-2}$ (19). Any nonaqueous alkali or alkaline earth metal system which shows electrode interfacial capacitance smaller than 5 $\mu\text{F cm}^{-2}$ should be suspected for an anode being coated by SEI. Scarr (20) measured 2.4–5.2 $\mu\text{F cm}^{-2}$ in the Li/PC + LiClO₄ system. Values smaller than 1 $\mu\text{F cm}^{-2}$ were measured for the Mg/SOCl₂, Mg²⁺, and the Li/SOCl₂, Li⁺ systems (21). The anodic transfer coefficient (α_a) for the lithium electrode in nonaqueous solution was found to be smaller than 0.5 which means that the Tafel slope (b) is larger than 120 mV. The results are summarized in Table II. Scarr (20) found a smaller α_a value where the "double layer" capacity was smaller: α_a was 0.25 where C_{dl} was 2.4 $\mu\text{F cm}^{-2}$, α_a was 0.5 where C_{dl} was 5 $\mu\text{F cm}^{-2}$. This can be explained by different thicknesses of the SEI in these two experiments. If we assume that the dielectric coefficient of the SEI is 10, then in the first

Table II. The kinetic parameters for the Li electrode in nonaqueous system

$\alpha_a^{(1)}$	$\alpha_a^{(2)}$	$b^{(3)}$ (mV)	System	Ref.
0.25		250	Li (Hg), DMSO, LiCl	(22)
	0.2		PC, LiAlCl ₄	(7)
0.13		450	PC, LiAlCl ₄	(5)
	0.5		PC, LiClO ₄	(20)
	0.25		PC, LiClO ₄	(20)
	0.25		PC, LiAlCl ₄	(1)
	0.17–0.52 ⁽⁴⁾		TC, LiAlCl ₄	(23)

⁽¹⁾ From Tafel equation.

⁽²⁾ From Equation [18a].

⁽³⁾ Formally calculated using $b = 60/\alpha_a$ (mV).

⁽⁴⁾ Depending on the thickness of the SEI.

experiment L was about 40 Å and in the second one only 20 Å.

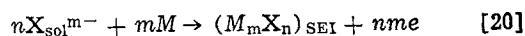
In other cases it was found that a linear relation between the overpotential and the current density is continuous beyond the expected value predicted by the Volmer-Butler equation. The lithium electrode in SOCl₂ shows ohmic behavior up to 600 mV (23). The Li electrode in pc exhibits ohmic behavior up to 200 mV (7). The value of the Tafel slope (b) should be at least five times larger than the highest overpotential value on the linear plot. Thus the expected Tafel slopes for these electrodes should be 3 and 1V (at least), respectively.

Complications associated with SEI batteries (SEI with $t_- > 0$).—So far the simplest SEI anodes having $t_+ = 1$ were discussed. These anodes are mostly desirable from a practical point of view. SEI anodes having $t_+ < 1$, $t_- > 0$ ($t_e = 0$) are undesirable for battery applications and may exhibit a much more complex behavior. The current through such SEI is carried both by cationic and anionic defects. As a result, on anodic polarization (discharge of the battery), anions will be injected from the solution into the SEI and will finally reach the M/SE interface. In such systems, especially where $t_- > t_+$, this process may lead to the increase of the thickness of the SEI and to increase in the polarization. In NAB having such SEI, the anode will be severely polarized and even blocked after a short period of discharge time.

Another disadvantage of such a system is the effect of t_- on the emf of the battery and on the anodic half-cell potential. These electrodes are not expected to follow a simple Nernst equation, but a more complicated one (19)

$$\Delta\phi_{M/sol} = t_+ \left(E^\circ - \frac{RT}{nF} \ln [M^{n+}]_{sol} \right) + \frac{t_-}{mF} (\Delta G \text{ side reaction}) \quad [19]$$

The over-all side reaction may be



where X^{m-} is an anion in the solution which is capable of migrating through the SEI.

It seems that this case is relevant to many NAB systems which use calcium or magnesium as the anode since in many magnesium or calcium compounds, the value of t_+ is smaller than 1 (15) (e.g., chlorides and oxides). This may be the reason we have no calcium or magnesium nonaqueous batteries.

It was found that calcium and magnesium SOCl₂ cells have less capacity than the lithium one (24–26). In addition, it was observed that the Ca anode in the SOCl₂ cell became coated by a white precipitate during discharge until it was completely blocked (25, 26).

A similar situation may exist in the Mg-MnO₂ dry cell (24). In this case the passivating layer on the surface of the anode (MgO or Mg(OH)₂) may have $t_{\text{Mg}^{+2}} < 1$. As a result the OCP of this cell is lower than expected. A similar phenomenon was observed during cycling of Li in LiBF₄-PC solutions (27). The stripping overpotential rose with time and a buildup of a passivating layer on the top of the lithium deposit was observed by SEM. This type of behavior may be attributed to SEI having $t_- > 0$. An attempt to improve the electrochemical performance of Mg electrode in SOCl₂ solution by doping the SEI with higher valance cations was made but with no success yet (24).

Shorts during deposition.—The plating current efficiency of lithium in organic systems reaches 100% (2, 14, 28, 27). However, the stripping efficiency is much lower than 100% and declines with wet standing and with the plating current density (27, 28). The reason for this phenomenon seems to be the formation of shorts through the SEI during plating as a result of local excessive heating and the high electric field on the SEI. The increase of the plating rate should aggravate the formation of shorts. The formation mechanism of these shorts may be the dissolution and trapping of metallic atoms in the cationic vacancies and of electrons in the anionic vacancies. This type of phenomenon occurs during the electrolysis of solid alkali halides at elevated temperatures and it causes the increase of the electronic conductivity of the solid (29). A more detailed description of this problem can be seen in Fig. 2. As a result of local heating, some part of the SEI becomes an electronic conductor (Fig. 2a), and therefore, lithium metal begins to deposit in the bulk of the SEI (Fig. 2b). This process continues and a ball of lithium totally covered by SEI is formed (Fig. 2c). As the current diminishes, this short zone in the SEI dis-

appears and later, when the battery is discharged, this insulated lithium ball will not be dissolved (Fig. 2d). This results in a decrease in the discharge current efficiency. In order to alleviate this problem the ionic resistivity of the SEI should be decreased by doping. This way both the electric field, which will be developed on the SEI, and the electrical heating of the SEI will be reduced.

The growth rate of the SEI.—A slow corrosion rate, and therefore, a long shelf life is expected for systems in which, at least, one of the corrosion products is very insoluble and immediately precipitates on the anode surface to form a dense and hole-free SEI which is a very good insulator for electrons. However, even in this case, the thickness of the SEI will slowly increase with time of storage. The rds for this process will always be the cathodic reaction, i.e., the reduction of the solvent. The functional relation between the thickness of the SEI and the storage time is determined by the mechanism of the cathodic reaction. Two extreme cases can be identified. (i) The anode surface and the SEI are not completely homogeneous, i.e., the anode contains anodic areas in which metallic ions are dissolved and cathodic areas in which the electrons are migrating through the SEI and reducing the solvent. This case may be relevant to practical systems having impurities which are able to create cathodic zones. (ii) In an ultrapure system, the anode and the SEI are completely homogeneous and thus no well-defined cathodic areas can exist. In this case, the rds will be the diffusion of electrons through the SEI to the solution side.

For the first case, the growth rate of the SEI can be formulated as follows

$$V(\text{corrosion}) = V_{\text{eq,cell}} - \eta_c - \eta_a - iR = 0 \quad [21]$$

$V_{\text{eq,cell}}$ = OCP ($i = 0$), η_c , η_a = cathodic and anodic overpotential; R = resistance of the solution. In practical NAB systems both η_a and iR can be neglected. Thus

$$\eta_c = V_{\text{eq,cell}} \quad [22]$$

If we assume that (i) the corrosion current follows Ohm's law and (ii) the electronic resistivity of the SEI [$\rho(e)$] is constant with time then we get

$$i_{\text{corr}} = \frac{V_{\text{eq,cell}}}{\rho(e) \cdot L} \quad [23]$$

It was assumed that all the corrosion products precipitate on the anode, to give a homogeneous film thus

$$\frac{dL}{dt} = Ki_{\text{corr}} \quad [24]$$

where K is a constant.

From Eq. [23] and [24] one gets

$$\frac{dL}{dt} = \frac{KV_{\text{eq,cell}}}{\rho(e) \cdot L} \quad [25]$$

After integration of Eq. [25]

$$L = \left(L_0^2 + \frac{2KV_{\text{eq,cell}}}{\rho(e)} \cdot t \right)^{1/2} \quad [26]$$

where $L_0 = L$ at $t = 0$. This is a parabolic law of growth of the SEI.

The second mechanism involves the diffusion of electrons through the SEI as the rds for the corrosion of the anode. The corrosion current density will follow Eq. [27]

$$i_{\text{corr}} = \frac{FDC_0}{L} \quad [27]$$

where D is the diffusion coefficient of the electrons in the SEI and C_0 is the concentration of the electrons in

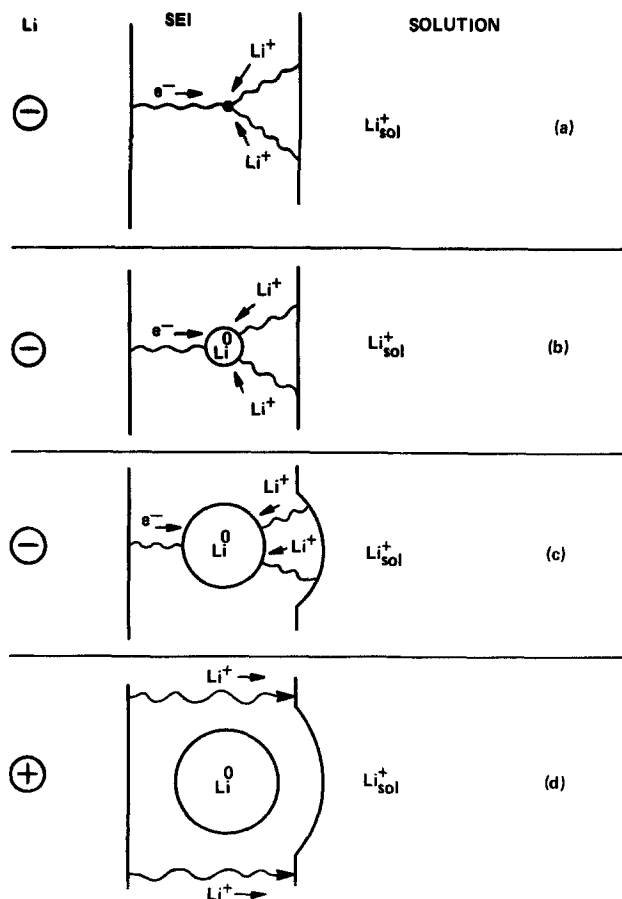


Fig. 2. The effect of a partial short on the deposition-dissolution process (see text).

the SEI near the M/SE interface. From Eq. [24] and [27]

$$\frac{dL}{dt} = \frac{KFDC_0}{L} \quad [28]$$

After integration a parabolic law of growth is obtained

$$L = (L_0^2 + 2KFDC_0 t)^{1/2} \quad [29]$$

Although the two corrosion mechanisms both lead to parabolic law of growth they should be considered as first approximation only. A practical system may be more complicated due to the following: $\rho(e)$ or D may change with L ; the SEI may contain cracks, holes, and grain boundaries at which the greatest part of the electronic leakage current may take place. The SEI may be nonhomogeneous and contain crystals of different sizes (21). Therefore deviations from the parabolic law of growth are expected.

The thickness of the SEI of Li in $\text{LiAlCl}_4\text{-SOCl}_2$ system was measured as a function of time at room temperature. The value of α in Eq. [30] was calculated for several Li electrodes (21)

$$L = \text{const} \cdot t^\alpha \quad [30]$$

Not all the Li electrodes exhibited the parabolic law of growth as values between 0.2 and 0.5 were obtained.

Summary

A model which is designed to explain the complex and unique electrochemical behavior of alkali and alkaline earth metals NAB systems was presented in this paper. This model is supported by many experimental results. It was indicated that the deposition-dissolution mechanism of alkali and alkaline earth metals in SEI-NAB systems is entirely different from that in aqueous or aqueous-like systems. Therefore the improvement or control of this process should be done by different means. It is concluded that a proper anodic SEI is the key for the operation of NAB. It seems that the controlling of the properties of the SEI, i.e., reducing t^- and t_e , increasing t^+ as close as possible to unity, and reducing ρ (by doping the SEI), will improve the performance of the SEI nonaqueous batteries.

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